

Education

Indian Institute of Technology Goa	2023 – Present
B.Tech in Mechanical Engineering	CGPA: 7.59/10

Publications & Preprints

UNITYAI-GUARD: Multilingual Toxicity Detection for Indic Languages [Website] [Paper] Rohit Kumar, et al.	EMNLP 2025, System Demos
StrategyIR: Risk-Calibrated Tactical Ranking over Cricket Action Spaces et al., Rohit Kumar, et al.	Preprint
FRAME-RAR: Frequency-Enhanced Motion History with Retrieval-Augmented Recognition for Real-Time Video Understanding Rohit Kumar	Preprint

Experience

Research Intern – IIT Patna [Supervisor: Dr. Sriparna Saha]	(Nov 2025 – Jan 2026)
- Contributed to the design of FieldSight, a dataset of 3,046 annotated ball-by-ball cricket instances with schema covering match context, field configuration, and tactical actions. - Implemented conformal risk control for uncertainty-aware tactical ranking, achieving 92% empirical coverage outperforming all baselines. - Built the player profiling module that aggregates retrieved evidence into uncertainty-aware coaching insights across 100 player profiles.	
Applied ML Lead – noCnX [Experience Letter]	(June 2025 – Dec 2025)
- Built MoldGPT, a multimodal RAG system for mechanical engineering SOPs, combining text and diagram retrieval with Whisper STT for multilingual voice queries on factory floors. - Designed the retrieval pipeline with document chunking, embedding-based semantic search, and context-aware answer generation using an LLM over proprietary internal data. - Integrated PostgreSQL and Redis for session management; built automated document ingestion pipeline with department-based access control.	
Research Intern – University of Waterloo [Verification Letter]	(June 2025 – Nov 2025)
- Conducted research on a multimodal AI approach for 3-month mRS prediction in acute ischemic stroke patients. - Investigated fusion strategies combining imaging biomarkers and clinical risk factors. - Evaluated ensemble ML models for stroke volume and clot burden assessment.	
Research Intern – IIT Gandhinagar [Supervisor: Dr. Mayank Singh]	(Feb 2025 – Apr 2025)
- Built a multilingual React + FastAPI platform with speech-to-text and transliteration support for toxicity classification across 6 low-resource Indic languages. - Curated the first large-scale Punjabi toxicity dataset (39k+ instances) addressing code-switching, transliteration, and cultural sensitivity across diverse online sources. - Fine-tuned mBERT for Punjabi toxicity classification achieving 80.29% accuracy on a manually verified test set of 5k instances.	

Projects

LearnWise : AI-Powered Lecture Comprehension Assistant [Code] [Documentation]	(Sept 2025 – Oct 2025)
- Built a multi-agent RAG platform transforming lecture videos and PDFs into structured notes, MCQ quizzes, and interactive Q&A using Whisper for transcription and a custom FAISS-based retrieval index. - Designed modular agents for summarization, quiz generation, and conversational Q&A powered by LLaMA-3 and Mistral, with model selection based on task complexity.	
Real-Time Object Detection Flutter App with Text-to-Speech [Code] [Presentation]	(Nov 2024 – Dec 2024)
- Built a Flutter application for real-time object detection using Google ML Kit and MobileNetV3, supporting 10K+ object categories. - Integrated text-to-speech to audibly announce detected objects, improving accessibility for visually impaired users.	
AI Based Tool to assist visitors at Police Station [Code] [Presentation]	(Sept 2024 – Oct 2024)
- Built two NLP chatbots, COPBOT (incident classification & case filing) and COPHELP (legal FAQ), with a bilingual voice-activated web UI supporting Hindi/English. - Created a custom legal dataset; trained TF-IDF + Logistic Regression models enhanced with Hugging Face Transformers; deployed via Flask with admin login and real-time workflows.	

Skills

Programming Skills:	Python, C++, Java, SQL, JavaScript, HTML, CSS, Kotlin, Dart
Libraries & Frameworks:	PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, HuggingFace, LangChain, FAISS, LoRA, PEFT, vLLM, React, Flask, FastAPI, Flutter
Tools & Platforms:	Git, Docker, SLURM, HPC/Linux, VS Code, Android Studio, FFmpeg, Ngrok
ML Techniques:	Conformal Prediction, GRPO, SFT, Multilingual NLP, Retrieval-Augmented Generation, Fine-tuning

Achievements

- Runner-up at Goa Police Hackathon 2024, securing 2nd place among 32 state-level teams.
- Qualified Indian Olympiad Qualifier in Mathematics (IOQM), clearing the first stage of the national mathematics olympiad.
- Qualified Kishore Vaigyanik Protsahan Yojana (KVPY) aptitude test, a national fellowship exam for science research.